

Scalable Fantasy Baseball Draft Optimization: Exact ADP-Aware ILP and Opportunity-Cost Greedy Heuristics

Group 4

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Abstract

We study fantasy baseball snake-draft roster construction as an operations research problem. The decision maker must select one player at each owned pick, assign drafted players to roster positions, and account for player availability implied by average draft position (ADP). We formulate an ADP-aware integer linear program (ILP) as an exact benchmark. Because this model becomes large as the player pool and roster size grow, we propose Opportunity Cost Greedy, a scalable heuristic that drafts where delaying to the next owned pick is most costly. Direct Greedy is included as a simple benchmark heuristic. Synthetic experiments evaluate point distributions, position distributions, ADP uncertainty, and large-scale stress settings, while 2026 Yahoo and FanGraphs data provide realistic validation. Opportunity Cost Greedy consistently has smaller optimality gaps than Direct Greedy and solves large synthetic instances in seconds.

1 Introduction

Fantasy Baseball is a strategic game in which participants act as team managers and build rosters using real Major League Baseball players. During the season, the real-life statistics of these players, such as home runs, stolen bases, strikeouts, and innings pitched, are converted into fantasy points by a scoring system. Before the season begins, most leagues allocate players through a draft. In a Head-to-Head points league, a good draft can create a large advantage because it determines the initial roster that generates most of the team’s expected season value.

This project focuses on the snake draft format. In a snake draft, managers take turns selecting players, and the order reverses every round. For example, in a 12-team league, the manager with the first pick in Round 1 receives the last pick in Round 2. This reversal makes the problem more interesting than simply ranking players: a manager may wait a long time between two consecutive picks, and the player pool can change substantially before the next decision.

Draft planning is challenging for several reasons. First, a valid fantasy roster must satisfy strict position requirements, such as filling catcher, infield, outfield, utility, starting pitcher, and relief pitcher slots. Second, many players have multi-position eligibility, which creates flexibility but also complicates assignment decisions. Third, the draft is a competitive environment. A manager cannot assume that a desirable player will remain available later, because other managers are simultaneously selecting from the same pool. Average Draft Position (ADP), estimated from market drafts and mock drafts, provides a practical proxy for this availability risk.

These observations create a real decision-making trade-off. Selecting the highest projected points at the current pick may lead to an unbalanced roster or force the manager into weak positions later. Waiting on a position may be attractive if replacement players are expected to remain, but costly if the position becomes

scarce before the next pick. The manager must therefore balance player value, positional scarcity, roster feasibility, and expected market availability. This is a natural setting for operations research: the exact optimization model can clarify the benchmark solution, while a scalable heuristic can provide practical decisions when the instance becomes too large for exact optimization.

Our study compares three methods. ADP-aware ILP is the exact benchmark that optimizes projected roster points under roster, eligibility, snake-pick, and ADP availability constraints. Direct Greedy is a simple benchmark heuristic that fills the scarcest open position first. Opportunity Cost Greedy is our proposed scalable heuristic, designed to select players where waiting until the next pick is most costly.

2 Problem Description

We consider the decision-making process of one fantasy manager participating in a snake draft. The manager knows the league size, draft position, roster rules, scoring system, and the list of remaining players. The manager also has forecasts for each player’s fantasy points and ADP, plus the positions that each player can fill. The planning task is to decide which player to select at each owned draft pick and how the selected players should be assigned to the final roster.

Conceptually, the manager controls two types of decisions. The first decision is selection: at each owned pick, exactly one player should be drafted from the currently available pool. The second decision is assignment: each drafted player must be placed into one roster slot that the player is eligible to fill. Multi-position players create useful flexibility, but they also create an assignment problem because using a flexible player in one slot may block a better roster configuration later.

The important inputs are player projections, ADP values, position eligibility, roster requirements, league size, draft order, and the ADP buffer used to represent availability uncertainty. Projected points measure the expected contribution of each player. ADP measures market availability: if a player is usually drafted much earlier than our current pick, we should not assume that the player is available. Roster requirements define the operational feasibility rules of the fantasy team, such as the number of outfielders, starting pitchers, and relief pitchers that must be filled.

The objective is to maximize the expected total fantasy points of the final roster. The constraints reflect business rules of the draft: each pick can select only one player, no player can be selected twice, every drafted player must fill exactly one roster slot, every roster slot requirement must be satisfied, players can only be assigned to eligible positions, and players whose ADP suggests that they should already be gone cannot be selected at that pick. In the exact benchmark, these decisions are optimized jointly. In the greedy heuristics, decisions are made sequentially to imitate a scalable draft assistant.

3 Model Formulation

Let I be the player set, P the roster-position set, and K the set of owned picks. Let V_i be projected points, $E_{ip} \in \{0, 1\}$ eligibility, R_p required slots, A_i ADP, and S_k the overall pick number. For a snake draft with M teams and draft position j , the pick in round r is

$$S_r = \begin{cases} (r-1)M + j, & r \text{ odd}, \\ rM - j + 1, & r \text{ even}. \end{cases}$$

The binary variables are y_i , whether player i is drafted; x_{ip} , whether player i is assigned to position p ;

and z_{ik} , whether player i is selected with owned pick k . The ADP-aware ILP is

$$\max \sum_{i \in I} V_i y_i \quad (1)$$

$$\text{s.t.} \quad \sum_{i \in I} z_{ik} = 1 \quad \forall k \in K, \quad (2)$$

$$\sum_{k \in K} z_{ik} = y_i \quad \forall i \in I, \quad (3)$$

$$\sum_{p \in P} x_{ip} = y_i \quad \forall i \in I, \quad (4)$$

$$x_{ip} = 0 \quad \forall i \in I, p \in P \text{ with } E_{ip} = 0, \quad (5)$$

$$\sum_{i \in I} x_{ip} = R_p \quad \forall p \in P, \quad (6)$$

$$z_{ik} = 0 \quad \forall i \in I, k \in K \text{ with } A_i + \delta < S_k, \quad (7)$$

$$x_{ip} \in \{0, 1\} \quad \forall i \in I, p \in P, \quad (8)$$

$$y_i \in \{0, 1\} \quad \forall i \in I, \quad (9)$$

$$z_{ik} \in \{0, 1\} \quad \forall i \in I, k \in K. \quad (10)$$

Constraint (2) selects one player at each owned pick. Constraint (3) links pick decisions to drafted-player decisions. Constraint (4) assigns each drafted player to one roster slot, while constraints (5) and (6) enforce eligibility and roster requirements. Constraint (7) removes player-pick pairs that are unavailable under the ADP buffer. The default roster has 16 slots: C, 1B, 2B, 3B, SS, three OF, Util, five SP, and two RP. The model is exact for the single-team planning problem under the ADP availability assumption. Its approximate variable count is $n(1 + p + r)$, where n is the number of players, p the number of roster positions, and r the roster size.

4 Algorithms

4.1 Direct Greedy: simple benchmark heuristic

Direct Greedy is intentionally simple. At each owned pick, it expires players whose $A_i + \delta$ is below the current pick. For each open position p , it computes

$$\text{scarcity ratio}_p = \frac{\text{remaining slots}_p}{\text{available eligible players}_p}.$$

It chooses the position with the largest scarcity ratio, breaking ties by the best currently available projected points, and drafts the highest-point available player for that position. It is fast and transparent, but it does not compare current value with future replacement value.

4.2 Opportunity Cost Greedy: proposed scalable heuristic

Opportunity Cost Greedy is the proposed scalable heuristic. It maintains two views of availability: players available at the current pick, and players expected to remain at the next owned pick. For each open position p , it computes

$$\text{opportunity cost}_p = \text{best current points}_p - \text{best next-pick points}_p.$$

The method drafts the player-position pair with the largest opportunity cost. It also includes a feasibility fallback. If the current or future eligible supply for a position is at or below the remaining required slots, that position receives forced priority:

$$\text{current count}_p \leq \text{remaining slots}_p \quad \text{or} \quad \text{future count}_p < \text{remaining slots}_p.$$

This prevents the heuristic from waiting too long in tight-supply cases.

Both heuristics use one max-heap per roster position, lazy deletion for expired or selected players, and active counts of available players by position. If e is the average number of eligible positions per player, the practical running time is approximately

$$O(ne \log n + rp \log n),$$

which is much smaller than constructing and solving the full ILP at large scale.

5 Data

5.1 Real data

The real-world validation instances use 2026 Yahoo and FanGraphs fantasy baseball player pools. Projection and ADP files are processed into player names, projected points, ADP values, and eligible positions. These datasets are realistic sanity checks: they show whether the synthetic findings also hold on actual player pools and scoring systems.

5.2 Synthetic data generation

Synthetic data are the main controlled experiment source. The base roster ratio follows the fantasy baseball roster above. Roster scale s multiplies all slot requirements, so $s = 1, 2, 3$ gives roster sizes 16, 32, 48. Let $D = rt$ be total league draft demand for roster size r and league size t . The player-demand ratio is n/D , with typical values 1, 3, 10. Points are generated independently from positions by default. ADP is generated from true point rank plus noise:

$$\text{ADP}_i = \text{rank}_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma_{\text{ADP}}),$$

then clipped to the player-pool range.

The point scenarios are normal, uniform, and high-low. The high-low scenario creates a small elite group and many low-value players, so missing an elite player is more costly. The position scenarios control eligibility and flexibility: uniform-by-type randomly samples hitter and pitcher patterns; point-flexible gives high-value players more eligibility; single-position removes flexibility; roster-ratio matches player position supply to roster demand.

6 Experiment Design

To systematically evaluate the effectiveness and robustness of our proposed heuristic algorithm, we conducted a comprehensive numerical study following the factorial design methodology, generating random instances designed with different factors and scenarios. The synthetic study evaluates algorithmic behavior under controlled environments, while the 2026 Yahoo and FanGraphs datasets provide real-world validation.

General Settings: Across the main synthetic instances, the base league consists of $t = 12$ teams and a roster size of $r = 16$. The roster requirements are one C, one 1B, one 2B, one 3B, one SS, three OF, one Util, five SP, and two RP. Let $D = rt$ be total league draft demand and n be the number of generated players. The default player-demand ratio is $n/D = 3$, so the base player pool has $D = 192$ total demanded roster slots and $n = 576$ players. Unless otherwise specified, the draft position is the middle draft slot, $\delta = 0$, $\sigma_{\text{ADP}} = 30$, and 10 random seeds are used.

Projected points are generated independently from positions by default. The normal point scenario samples from a clipped normal distribution with mean 400, standard deviation 120, and support $[10, 800]$. The uniform scenario samples $U(10, 800)$. The high-low scenario assigns 10% of players to an elite group sampled from $U(500, 800)$, while the remaining 90% are sampled from $U(10, 500)$. ADP is generated by ranking players by projected points, adding normal noise $N(0, \sigma_{\text{ADP}})$, and clipping the result to $[1, n]$.

We designed six scenario groups driven by five operational factors to test the algorithms under various draft environments:

- **Factor A: Player Value Distribution.** We vary the distribution of projected fantasy points to test whether heuristic performance changes when player value is smooth or concentrated. The *Normal* scenario is the baseline balanced market. The *Uniform* scenario creates a broad value spread. The *High-Low* scenario creates a small elite group and many low-value players, making missed elite picks more costly.
- **Factor B: Position Eligibility Structure.** We vary eligibility generation to test roster flexibility and positional scarcity.
 - **Uniform By Type:** Players are first classified as hitters or pitchers using roster proportions $P(H) = 9/16$ and $P(P) = 7/16$. Hitter and pitcher eligibility patterns are then sampled from predefined position sets.
 - **Point Flexible:** High-value players are more likely to receive multiple eligible positions. This represents a favorable setting where valuable players are easier to fit into the roster.
 - **Single Position:** Every player has exactly one eligible position. This is the lowest-flexibility setting and should make greedy mistakes harder to repair.
 - **Roster Ratio:** Primary positions are sampled in proportion to roster demand; for example, OF appears about three times as often as C because the roster has three OF slots and one C slot.
- **Factor C: Roster and Player-Pool Scale.** We vary the roster scale $s \in \{1, 2, 3\}$ and player-demand ratio $n/D \in \{1, 3, 10\}$. Scaling s multiplies every roster requirement, so $s = 1, 2, 3$ correspond to roster sizes $r = 16, 32, 48$. Larger n and r directly increase the number of ILP variables $n(1 + p + r)$.
- **Factor D: ADP Availability Uncertainty.** We vary $\sigma_{\text{ADP}} \in \{0, 10, 30, 60, 100\}$ and $\delta = -10, \dots, 10$. When $\sigma_{\text{ADP}} = 0$, ADP perfectly follows projected points. Larger values create more noisy availability, including both bargains and traps.
- **Factor E: Large-Scale Stress.** We combine high-low points, single-position eligibility, $\sigma_{\text{ADP}} = 60$, larger roster scales, larger league sizes, and larger player pools. These scenarios are intentionally constructed to test whether the exact ADP-aware ILP becomes slow or impractical, while the heuristics still produce feasible rosters quickly.

The parameter configurations are summarized by factor-level scenarios in Tables 1–3. The labels in these tables describe the experimental condition itself rather than the output-folder names, so the reader can see exactly which operational factor is being changed in each scenario.

Table 1: Factor-level synthetic scenarios for baseline, point, position, and ADP experiments.

Scenario	Main factor	Varying setting	Fixed controls	Cases
S0	Baseline	normal points, roster-ratio positions, $\sigma_{\text{ADP}} = 30, \delta = 0$	$s = 1, t = 12, n/D = 3$	10
A1	Player value distribution	uniform points	roster-ratio positions, $s = 1, t = 12, n/D = 3, \sigma = 30, \delta = 0$	10
A2	Player value distribution	high-low points	roster-ratio positions, $s = 1, t = 12, n/D = 3, \sigma = 30, \delta = 0$	10
B1	Position eligibility	uniform-by-type positions	normal points, $s = 1, t = 12, n/D = 3, \sigma = 30, \delta = 0$	10
B2	Position eligibility	point-flexible positions	normal points, $s = 1, t = 12, n/D = 3, \sigma = 30, \delta = 0$	10
B3	Position eligibility	single-position only	normal points, $s = 1, t = 12, n/D = 3, \sigma = 30, \delta = 0$	10
B4	Position eligibility	roster-ratio positions	normal points, $s = 1, t = 12, n/D = 3, \sigma = 30, \delta = 0$	10
D1	ADP uncertainty	$\sigma_{\text{ADP}} = 0, \delta = -10, \dots, 10$	normal points, roster-ratio positions, $s = 1, t = 12, n/D = 3$	210
D2	ADP uncertainty	$\sigma_{\text{ADP}} = 10, \delta = -10, \dots, 10$	normal points, roster-ratio positions, $s = 1, t = 12, n/D = 3$	210
D3	ADP uncertainty	$\sigma_{\text{ADP}} = 30, \delta = -10, \dots, 10$	normal points, roster-ratio positions, $s = 1, t = 12, n/D = 3$	210
D4	ADP uncertainty	$\sigma_{\text{ADP}} = 60, \delta = -10, \dots, 10$	normal points, roster-ratio positions, $s = 1, t = 12, n/D = 3$	210
D5	ADP uncertainty	$\sigma_{\text{ADP}} = 100, \delta = -10, \dots, 10$	normal points, roster-ratio positions, $s = 1, t = 12, n/D = 3$	210

Table 2: Moderate scale scenarios, crossing roster scale and player-pool size.

Scenario	Roster scale s	Roster size r	Teams t	n/D	Players n
C1	1	16	12	1	192
C2	1	16	12	3	576
C3	1	16	12	10	1,920
C4	2	32	12	1	384
C5	2	32	12	3	1,152
C6	2	32	12	10	3,840
C7	3	48	12	1	576
C8	3	48	12	3	1,728
C9	3	48	12	10	5,760

Table 3: Large-scale stress ladder.

Level	Roster scale s	Roster size r	Teams t	n/D	Players n
E1 stress_small	3	48	12	10	5,760
E2 stress_medium	4	64	15	10	9,600
E3 stress_large	6	96	15	15	21,600
E4 stress_xlarge	10	160	20	20	64,000
E5 timeout_target	16	256	24	30	184,320

The performance metrics are objective value, runtime, method status, Gurobi MIPGap for ILP, and heuristic optimality gap. For a heuristic solution H , the optimality gap is

$$\text{gap}(H) = \frac{\text{objective}_{\text{ILP}} - \text{objective}_H}{\text{objective}_{\text{ILP}}}.$$

This gap is computed only against ADP-aware ILP, not against a no-ADP model.

7 Results

7.1 Baseline, points, and positions

Table 4 shows the main quality results for the baseline, point-distribution, and position-eligibility scenarios. These experiments are designed to explain where heuristic errors come from before moving to larger scaling experiments.

Table 4: Heuristic quality in baseline, point-distribution, and position-distribution experiments.

Scenario	ILP mean objective	Opportunity Cost gap	Direct Greedy gap
S0 baseline	9228.582	1.919%	4.785%
A1 uniform points	11440.292	1.439%	3.243%
A2 high-low points	9137.382	2.867%	8.390%
B1 uniform-by-type	9112.331	1.074%	4.889%
B2 point-flexible	9139.980	1.344%	5.364%
B3 single-position	9070.910	0.939%	5.107%
B4 roster-ratio	9228.582	1.919%	4.785%

Opportunity Cost Greedy is consistently closer to the ADP-aware ILP than Direct Greedy. The high-low point distribution is the hardest point scenario because missing elite players is expensive. Position distribution does not create infeasibility after adding the feasibility fallback, but Direct Greedy remains more sensitive to eligibility structure.

7.2 ADP uncertainty

Scenarios D1–D5 vary ADP noise and the availability buffer. Table 5 aggregates over all δ values for each noise level. As ADP becomes noisier, both heuristics generally move farther from the exact benchmark, but Opportunity Cost Greedy remains substantially closer.

Table 5: ADP uncertainty experiment. Each row aggregates 210 cases.

σ_{ADP}	ILP mean objective	Opportunity Cost gap	Direct Greedy gap
0	8449.595	0.416%	1.721%
10	8690.528	0.992%	3.342%
30	9243.851	1.800%	5.029%
60	9761.266	2.429%	5.728%
100	10047.434	2.146%	4.496%

7.3 Moderate and stress scaling

The moderate scale grid C1–C9 confirms that both heuristics remain close to the exact benchmark while using much less runtime. Across 45 moderate scale cases, ADP-aware ILP has mean runtime 0.644 seconds and max runtime 4.915 seconds. Opportunity Cost Greedy has mean gap 1.623%, mean runtime 0.093 seconds, and max gap 3.714%. Direct Greedy has mean gap 4.177%, mean runtime 0.088 seconds, and max gap 10.349%.

The stress ladder E1–E5 then extends to much larger instances. Table 6 shows the scale separation. The ILP remains exact when it completes, but runtime grows sharply. At the timeout-target size, the ILP run

was manually interrupted without producing benchmark output, while both heuristics solved the instances in under ten seconds.

Table 6: Runtime scaling on synthetic stress instances.

Instance	Approx. variables	ADP-aware ILP	Opportunity Cost	Direct Greedy
stress_large	2,289,600	41.131s	1.008s	0.926s
stress_xlarge	10,880,000	299.430s	2.965s	2.705s
timeout_target	49,029,120	interrupted	9.381s	8.744s

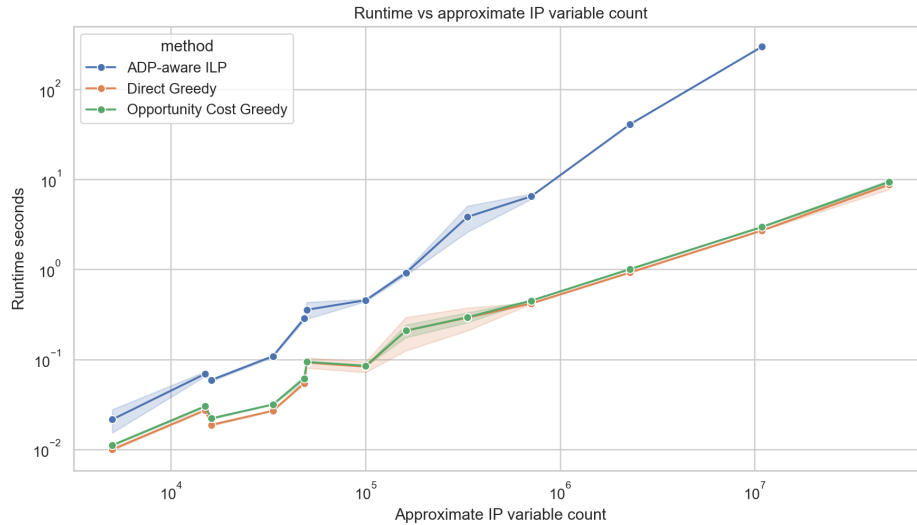


Figure 1: Runtime grows sharply for the exact benchmark, while both heuristics remain fast on large synthetic instances.

7.4 Real-data validation

The 2026 Yahoo and FanGraphs benchmarks are realistic validation cases rather than the main experiment grid. Each dataset has 252 benchmark cases. Table 7 again shows that Opportunity Cost Greedy has smaller gaps than Direct Greedy.

Table 7: 2026 real-data validation summary.

Dataset	Method	Mean objective	Mean gap	Max gap
Yahoo 2026	ADP-aware ILP	7917.209	0.000	0.000
Yahoo 2026	Opportunity Cost Greedy	7767.732	149.477	255.700
Yahoo 2026	Direct Greedy	7662.866	254.343	373.000
FanGraphs 2026	ADP-aware ILP	14642.927	0.000	0.000
FanGraphs 2026	Opportunity Cost Greedy	14320.135	322.792	508.000
FanGraphs 2026	Direct Greedy	14097.337	545.590	644.500

8 Conclusions

The ADP-aware ILP provides a precise exact benchmark for fantasy baseball draft optimization on manageable instances. Direct Greedy is useful as a simple benchmark heuristic, but it often leaves more value on the table. Opportunity Cost Greedy, our proposed scalable heuristic, captures the value of drafting before a position deteriorates and includes a fallback for tight feasibility. Across synthetic experiments and real-data validation, it consistently improves over Direct Greedy while remaining fast.

The main implication is that exact optimization is valuable for benchmarking and small to medium instances, while a carefully designed heuristic is necessary for large-scale draft planning. Future work could add opponent-specific draft behavior, use calibrated scoring variants as robustness checks, and combine the heuristic with small local ILP subproblems for dynamic in-draft replanning.